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First UAE Deployment of At-The-Bit Steerable System in Unconventional Project: Enhancing Build Capability and Drilling Efficiency in High-DLS Applications

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Abstract

This abstract presents the first successful deployment of an At-the-Bit Rotary Steerable System (RSS) in the UAE, targeting an unconventional project. The goal was to address challenges in drilling an 8.5" production hole section with high dogleg severity (DLS) demands. These high-DLS wells aimed to reduce the Landing Point (LP) vertical section, minimizing the loss of potentially productive acreage in unconventional zones and optimizing overall well trajectory length while maintaining borehole quality and drilling efficiency.

The At-the-Bit RSS, equipped with dual hydraulically actuated pistons adjacent to the cutting structure, was selected to meet trajectory demands of up to 4°/100 ft DLS. A comprehensive risk assessment and Management of Change (MOC) process preceded deployment. The system was run with a 5-bladed bit featuring 16 mm cutters. The Bottom Hole Assembly (BHA) included shock and vibration suppression tools and was optimized for directional responsiveness. Continuous real-time monitoring was conducted to track tool performance and directional control.

The At-the-Bit RSS demonstrated strong directional performance, achieving build rates between 6–11°/100 ft, confirming its effectiveness in high-curvature drilling environments. The revised directional plan contributed to improved drilling efficiency by shortening the total depth by 150 ft, thereby reducing the negative vertical section and enhancing well placement. Operationally, the system exhibited smooth pull-out-of-hole behavior with low friction factors (0.2), indicating good borehole quality. However, elevated shock and vibration levels were recorded, which partially constrained tool performance. A follow-up deployment in a more complex section has been completed, confirming the system's adaptability to varied lithologies and its ability to maintain trajectory accuracy with minimal corrective action. For future deployments, especially in second or third trials—are expected to further validate tool building capability. Recommendations include using less aggressive bit designs to reduce vibration, continued use of vibration suppression tools, and trajectory designs optimized around a 6°/100 ft DLS.

This deployment marks the first use of an At-the-Bit RSS in the UAE, expanding the portfolio of high-performance directional drilling technologies in the region. The at-the-bit design with dual pistons enabled responsive steering in high-curvature environments without additional hydraulic force. The deployment also set a precedent for broader use of this technology in unconventional assets and contributed new learnings on bit/tool interaction under high DLS loads. The tool's benefits are expected to support future unconventional well campaigns through reduced LP vertical sections and shorter well lengths.

Introduction

The development of unconventional oil resources in the United Arab Emirates (UAE) has emerged as a strategic priority in operating company's long-term energy roadmap. As conventional reservoirs mature, operating company has taken the lead in advancing technologies and strategies to unlock tight formations that demand innovative drilling solutions (Al-Kaabi et al. 2021).

Among its flagship initiatives, the operating company project stands out as a pioneering effort targeting the Cretaceous Shilaif formation—a low-permeability reservoir located in the southern region of Abu Dhabi. Exploration efforts in structurally complex zones such as the Falaha and Ghurab Basins, and the Huwaila and Dhafra highs, have confirmed the presence of substantial hydrocarbon volumes. However, the geological nature of the Shilaif formation, with its tight pay intervals and limited natural permeability, presents significant challenges for conventional drilling approaches (Gulfam et al. 2020).

To economically access these resources, operating company has prioritized horizontal drilling as a core strategy. The goal is not only to reach the reservoir but to maximize contact with production zones while maintaining borehole stability and operational efficiency. In conventional well planning, the build-up to the landing point typically involves dogleg severities (DLS) of 3.0 to 3.5°/100 ft. While this trajectory is achievable with traditional rotary steerable systems (RSS), it often results in extended curve sections and deeper build trajectories. These "belly" profiles increase the total measured depth (MD), reduce vertical efficiency, and introduce risks such as wellbore instability, poor hole cleaning, and increased friction—especially in the curve interval. In unconventional reservoirs, where precise geosteering and minimal closure distance are critical, such limitations can compromise well productivity and increase costs (El-Gendy et al. 2018).

Recognizing these constraints, Operating company initiated a technology pilot in collaboration with SLB to evaluate the feasibility of higher-DLS trajectories. The objective was to reduce the extent of the belly profile, shorten total MD, and improve reservoir entry while avoiding undrained zones. This required departure from conventional RSS tools and the adoption of a more responsive steering mechanism capable of delivering build rates in the range of 5 to 6°/100 ft. The pilot aimed to validate whether such aggressive curvature could be achieved without compromising borehole integrity or tool reliability (Al Hosani et al. 2022).

Rotary steerable systems have transformed the directional drilling landscape since their introduction in the late 1990s. These tools allow continuous rotation of the drill string while steering the wellbore, offering significant advantages in rate of penetration (ROP), borehole quality, and trajectory control. Early RSS designs were based on two primary mechanisms: push-the-bit and point-the-bit. Push-the-bit systems use pads that extend from the tool body to push the bit sideways, while point-the-bit systems tilt the bit itself to change direction. Both approaches improved directional control compared to conventional motor assemblies, but each had limitations in high-curvature applications (Smith et al. 2019).

First-generation push-the-bit RSS tools were effective in conventional reservoirs but struggled to deliver aggressive build rates in unconventional plays. Their mechanical limitations and delayed steering response made them unsuitable for tight curvature requirements. Second-generation systems introduced improved toolface control and more robust mechanical designs, enabling build rates up to 3.5°/100 ft in many applications. However, even these tools faced challenges when confronted with the demands of high-DLS trajectories in complex formations like the Shilaif.

In the operating company project, the original well plan employed a second-generation push-the-bit RSS paired with a short-gauge bit. This configuration achieved a DLS of 3.5°/100 ft, which was sufficient for initial objectives. However, to further optimise well placement and reduce total depth, a revised directional plan was introduced. This new plan required build rates of 4°/100 ft, with several checkpoints demanding up to 6°/100 ft—exceeding the capabilities of conventional RSS tools.

To meet these enhanced trajectory demands, operating company and SLB selected an At-the-Bit Rotary Steerable System (RSS) for deployment—marking the first use of this technology in the UAE. Unlike traditional RSS tools that apply steering forces further up the Bottom Hole Assembly (BHA), the At-the-Bit RSS places dual hydraulically actuated pistons directly adjacent to the cutting structure. This design enables immediate curvature response, improved toolface control, and enhanced trajectory fidelity without the need for additional hydraulic force. These features make the tool particularly well-suited for high-DLS applications in unconventional reservoirs.

The deployment in the operating company project was preceded by a comprehensive risk assessment and Management of Change (MOC) process to ensure tool readiness and operational safety. The BHA was optimised with shock and vibration suppression tools to protect Measurement While Drilling (MWD) components and maintain tool integrity under high DLS loads. A 5-bladed bit with 16 mm cutters was selected to balance aggressiveness with stability, and real-time monitoring systems were employed to track tool performance, directional response, and borehole conditions throughout the run.

This introduction sets the stage for a detailed analysis of the engineering solutions applied, the performance metrics achieved, and the broader implications for future field development strategies in operating company's unconventional asset base. It also highlights the evolution of RSS technology and its growing role in enabling complex well trajectories that were previously considered unfeasible. A subsequent deployment in operating company's unconventional wells, conducted under more aggressive conditions, further demonstrated ABSS's success, reinforcing its reliability and expanding its applicability across challenging environments.

At the Bit Steerable System (ABSS): A Technological Leap

The collaboration between operating company and SLB on the field trial of the ABSS marks a significant advancement in directional drilling for unconventional reservoirs in the UAE. This trial was the first deployment of the ABSS tool in the country, representing a pivotal milestone in operating company's efforts to expand its unconventional drilling capabilities.

The ABSS (Fig. 1) is engineered for high-curvature drilling and offers enhanced directional control across complex 3D well profiles. Its standout feature is the dual hydraulically activated pistons positioned adjacent to the cutting structure. This design enables greater curvature leverage without requiring additional hydraulic pressure, allowing the tool to seamlessly transition between aggressive curve sections and smooth lateral drilling (Jain et al. 2021; He et al. 2022).

Figure 1—ABSS Components

The redesigned bias unit in the ABSS tool introduces several innovations:

- **Piston force actuators** for precise directional control.
- **Enhanced sealing mechanisms** that prevent fluid bypass, improving hydraulic efficiency.
- **Elimination of pad hinge points**, removing mechanical restrictions and improving reliability.

These enhancements collectively deliver:

1. **Higher and More Consistent Dogleg Severity (DLS)**: The tool maintains performance across varying formations, ensuring consistent trajectory control.
2. **Reduced Tortuosity and Improved Borehole Quality**: Dynamic force adjustments result in smoother curves and better wellbore integrity.
3. **Lower Risk of Well Deviation**: Real-time adaptability helps minimise deviations from the planned trajectory, reducing operational risks (Smith et al. 2020; Zhou et al. 2021).

The pilot deployment was conducted in the 8.5" production hole section, a critical part of the well that directly influences frack string runnability, coil tubing access, and long-term reservoir performance. Two case studies were executed:

- **Case Study 1**: A pilot run under moderate DLS conditions to assess the tool's baseline performance.
- **Case Study 2**: A validation run involving a more aggressive directional plan, aimed at benchmarking the tool's capabilities under challenging conditions.

Both trials demonstrated the ABSS's ability to deliver precise steering, high DLS, and improved borehole quality, reinforcing its value for operating company's unconventional portfolio.

This successful collaboration not only validates the ABSS's performance but also sets the stage for broader adoption of advanced RSS technologies in the region, aligning with operating company's strategic goals for innovation and efficiency in complex drilling environments. Let me know if you'd like help drafting a technical summary or presentation based on this.

Case Study 1 – Pilot Run with Moderate DLS

Challenge

Well "A", one of five wells located on a shared pad, was selected as the pilot candidate for deploying the ABSS tool. The original directional plan for this well featured a dogleg severity (DLS) of 3.5°/100 ft, which had previously been executed successfully using the 2nd Generation RSS paired with an AccuStrike bit. For the ABSS trial, the trajectory was revised to target a DLS of 4°/100 ft, with specific checkpoints reaching up to 6°/100 ft (Table 1). This adjustment was made to rigorously evaluate the tool's steering responsiveness and control in high-curvature drilling scenarios typical of unconventional environments.

Table 1—Case study 1 Direction Plan with 6°/100 ft.

Measured Depth (MD)	Course Length (CL)	Inclination (°)	Azimuth (°)	True Vertical Depth (TVD)	North-South (ft)	East-West (ft)	Vertical Section (ft)	Dogleg Severity (°/100 ft)	Toolface (°)	Build Rate (°/100 ft)	Turn Rate (°/100 ft)
6275	—	9.55	184	6210.1	-742.3	-27	-541.6	0	0	0	0
6542.3	267.3	10.47	290.63	6475.1	-756	2.5	-533.9	6	140.91	0.35	39.89
7582.7	1040.4	10.47	290.63	7498.4	-689.3	-174.5	-361.2	0	0	0	0
9549	1966.4	87.76	319.99	9697.9	283.3	1128.1	1001.2	4	29.08	3.93	14.49
16026.2	6477.1	87.76	319.99	8950.9	5241	5289.5	7446.2	0	0	0	0

Solution

To meet the revised directional objectives, the ABSS CL system was deployed with a 5-bladed PDC bit featuring 16 mm cutters. This bit was selected for its moderate aggressiveness, aimed at reducing the shock and vibration (S&V) levels commonly encountered in unconventional formations. To further safeguard the downhole tools, the SLB Sentinel suppressor was strategically placed in the bottomhole assembly (BHA) (Fig. 2) between the ABSS tool and the MWD realtime system. Acting as a damping mechanism, the Sentinel effectively absorbed axial shocks and mitigated vibration, thereby protecting the MWD components and ensuring the integrity of real-time data acquisition. This configuration enabled stable tool performance and reliable directional control throughout the run.

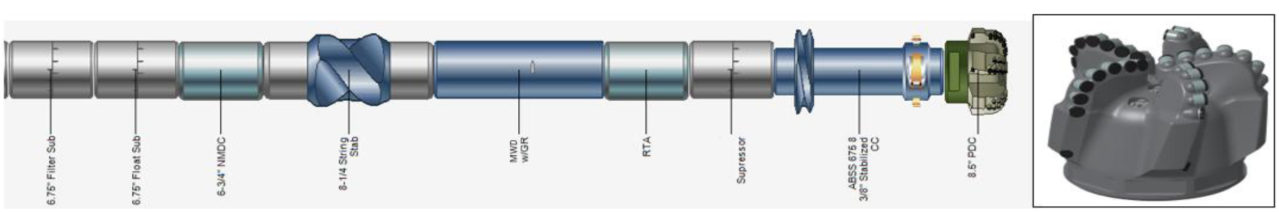

Figure 2—Case Study-1 First ABSS BHA with 516 PDC bit

Results

The pilot run of the ABSS was executed successfully, with the entire 8.5" production hole section drilled shoe-to-shoe. This encompassed approximately 10,000 ft of drilled footage, covering vertical, curve, and lateral segments. The lateral section alone measured around 6,000 ft, aligning precisely with contractual requirements and confirming the tool's suitability for extended reach applications in unconventional reservoirs.

Key Outcomes

Directional Control. The ABSS demonstrated robust steering performance throughout the curve section, achieving build rates between 5° and 7°/100 ft while operating at only 50–70% of its available steering force. This confirmed the tool's ability to manage both moderate and aggressive curvature profiles with precision. Toolface stability was maintained across the section, enabling consistent trajectory control and minimizing deviations.

As the section transitioned toward the lateral. However, geological changes DLS values tapered smoothly using 6°/100 ft, indicating controlled build-down and minimal microdoglegs. The following (Table 2) summarizes the curve section trajectory:

Table 2—Case study 1 Direction Control summary within Curve Section

MD (ft)	Inc (°)	Azim Grid (°)	TVD (ft)	DLS (°/100ft)
7693.00	14.75	297.04	7603.94	4.07
7788.00	20.69	300.97	7694.39	6.37
7880.00	23.02	303.90	7779.78	2.80
7974.00	25.25	307.72	7865.56	2.90
8066.00	28.39	311.70	7947.66	3.93
8158.00	31.50	313.34	8027.37	3.50
8251.00	35.49	313.09	8104.91	4.29
8344.00	39.36	312.24	8178.75	4.20
8436.00	41.55	313.38	8248.75	2.51
8531.00	47.44	311.32	8316.49	6.38
8622.00	52.17	313.01	8375.21	5.39
8714.00	56.72	315.12	8428.69	5.29
8807.00	61.17	317.14	8476.66	5.13
8900.00	65.54	318.61	8518.36	4.91
8992.00	68.21	318.54	8554.49	2.90
9085.00	71.22	318.78	8586.73	3.25
9177.00	74.07	318.84	8614.17	3.10

This data confirms the ABSS system's ability to deliver precise steering across varying formation conditions, supporting its deployment in high-curvature unconventional wells

Operational Efficiency. The revised well plan, enabled by ABSS high dogleg severity (DLS) capability, led to a 150 ft reduction in total depth (TD). This was achieved by deepening the kickoff point and shortening the curve section, which streamlined the trajectory (Fig. 3) and reduced drilling time. The optimised path not only improved rig efficiency but also lowered formation exposure and tool fatigue, contributing to safer and more cost-effective operations.

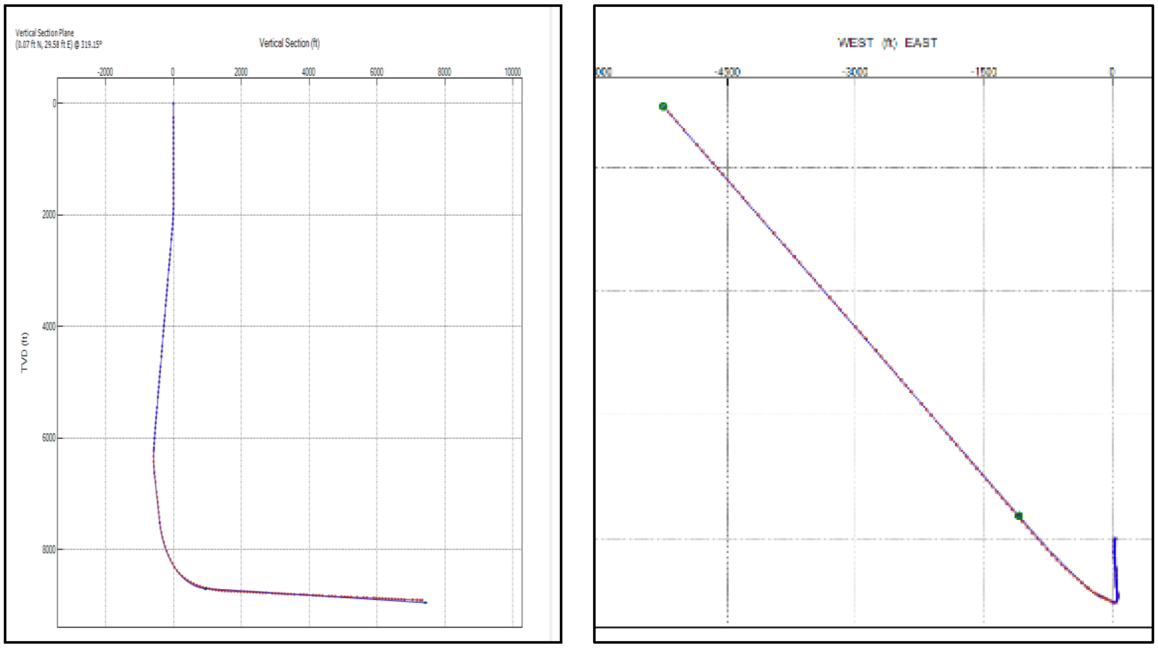

Figure 3—Well A Final Direction Survey Section and Plan view

Borehole Quality. The ABSS system delivered a high-quality wellbore, outperforming previous rotary steerable systems and motor BHAs used in similar formations. The curve and lateral sections exhibited minimal microdoglegs, contributing to a smooth trajectory. During POOH, a low friction factor of 0.2 (Fig. 4) was recorded in the open hole, indicating excellent borehole condition, and enhanced runnability for completion operations.

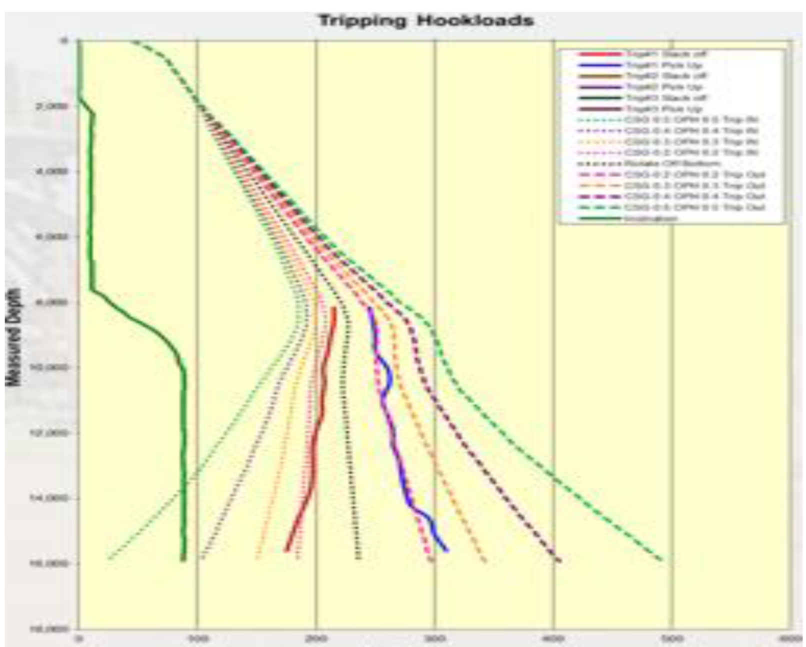

Figure 4—Last 8.5" Direction BHA Tripping loads chart while POOH (showing 0.2 FF)

Performance Limitation (ROP). While the ABSS system delivered strong directional control, the overall rate of penetration (ROP) was moderately limited due to elevated shock and vibration encountered during the run (Fig. 5). These dynamic loads were primarily caused by the high compressive strength of the carbonate formation, estimated between 15–20 ksi. Supporting data, including low gamma ray readings and minimal shale content, confirmed the challenging lithology. Vibration monitoring indicated significant lateral activity, which affected drilling stability and efficiency. To minimise the impact on the Measurement While Drilling (MWD) system, the SLB Sentinel suppressor was deployed in the bottomhole assembly. It helped preserve data quality and tool integrity, though formation hardness remained the dominant factor influencing ROP. This outcome highlights the importance of optimising bit design and managing dynamic loads when drilling in high-strength rock environments.

Figure 5—SnV Charts for Well A Using 8.5" BHA

Case Study 1 highlights the initial deployment of ABSS in Well "A" under moderate dogleg severity conditions. The tool achieved build rates up to 7°/100 ft with stable toolface control, reducing total depth by 150 ft and delivering excellent borehole quality. For future trials, it is recommended to use a less aggressive bit, including a vibration suppressor, and select a well requiring higher DLS to further validate performance.

Case Study 2: More Aggressive Directional Plan

Challenge

Following the initial pilot deployment of the ABSS, which aimed to validate tool performance under moderate conditions, the second run was initiated as a critical response to an operational challenge. Well "B" was selected for this trial—an exploratory well designed with a pilot hole and a lateral section. The pilot hole was successfully drilled, logged, and securely plugged back with cement, setting the stage for a sidetrack operation. This sidetrack was executed using a conventional motor-TCI bit assembly and proceeded without complications.

To drill the lateral section, a rotary steerable bottomhole assembly (Fig. 6) was mobilised, comprising the 2nd Generation RSS tool, an inline flex sub, and a 5-blade 16 mm Accustrike PDC bit. The flex sub was intended to enhance curvature delivery and help achieve the planned dogleg severity (DLS) of 4.4°/100 ft. However, the formation presented significant challenges. The lithology was highly variable, with interbedded soft and hard shale layers and abrasive limestone, resulting in pronounced anisotropy and unpredictable directional behaviour.

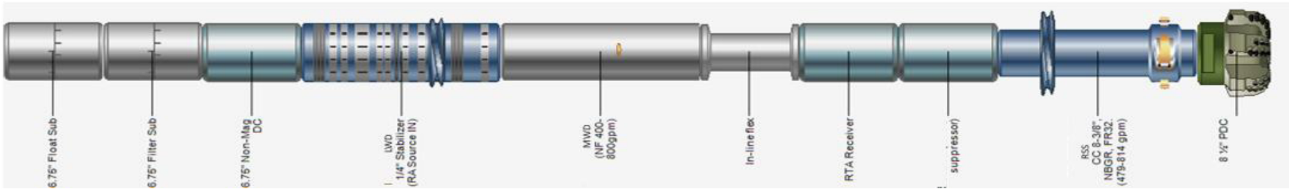

Figure 6—Case Study-2 BHA no 1: 2nd Generation RSS with Inline Flex Sub

Initially, the 2nd Generation RSS tool showed reasonable directional responsiveness, but as drilling progressed, it struggled to maintain consistent weight transfer and directional stability. The system plateaued at a maximum DLS of approximately 3.5°/100 ft (Table 3), falling short of the planned trajectory. This limitation became increasingly critical as the BHA approached the landing point—roughly 1,000 ft away—where sharper curvature was required to remain within the target TVD window.

Table 3—Case Study 2 BHA no 1: Deviation Survey

MD (ft)	Inc (°)	Azim Grid (°)	TVD (ft)	DLS (°/100ft)
7767.00	6.42	113.88	7766.07	5.56
7864.00	5.15	94.67	7862.58	2.37
7963.00	4.71	82.95	7961.22	1.11
8057.00	4.84	109.56	8054.90	2.34
8144.00	4.88	112.17	8141.59	0.26
8247.00	4.79	116.55	8244.22	0.37
8330.00	5.37	119.91	8326.90	0.79
8424.00	6.75	121.20	8420.37	1.48
8519.00	7.88	115.62	8514.60	1.40
8613.00	10.55	112.63	8607.38	2.88
8707.00	15.12	119.17	8699.01	5.09
8799.00	18.91	125.45	8786.97	4.57
8891.00	20.33	127.31	8873.63	1.69
8983.00	22.58	130.14	8959.25	2.69
9076.00	26.19	132.60	9043.94	4.03
9170.00	30.72	135.68	9126.57	5.06
9263.00	35.01	140.07	9204.68	5.27
9355.00	37.99	141.42	9278.63	3.35
9448.00	40.72	140.39	9350.53	3.02
9543.00	43.69	139.53	9420.89	3.18

Compounding the issue were severe stick-slip vibrations and borehole spiraling, which further degraded drilling efficiency and borehole quality. Despite multiple corrective attempts, including adjustments to drilling parameters over three stands, the trajectory could not be recovered. With only 700 ft remaining to reach the landing point and the required DLS increasing, the team made the decision POOH to reassess the BHA configuration and avoid further deviation from the planned path.

This experience highlighted the limitations of conventional push-the-bit RSS tools in high-curvature, anisotropic formations. It underscored the need for more advanced steerable systems capable of delivering higher DLS with greater stability and adaptability.

Solution

To achieve the well target and avoid a sidetrack, a revised trajectory plan was introduced requiring a dogleg severity (DLS) exceeding 5°/100 ft. To meet this build-rate, the ABSS was deployed (Fig. 7), based on its proven performance in Well A. The same bottomhole assembly (BHA) configuration was retained, including the NT516 bit, with the only change being the addition of the Ecoscope logging-while-drilling (LWD) tool due to the exploratory nature of Well B—though its inclusion introduced additional operational risk, particularly related to source management and tool reliability in high-angle drilling environments. The tool's dual hydraulic pads, positioned adjacent to the cutting structure, provided enhanced curvature control and steering leverage, which proved especially effective in constrained build windows.

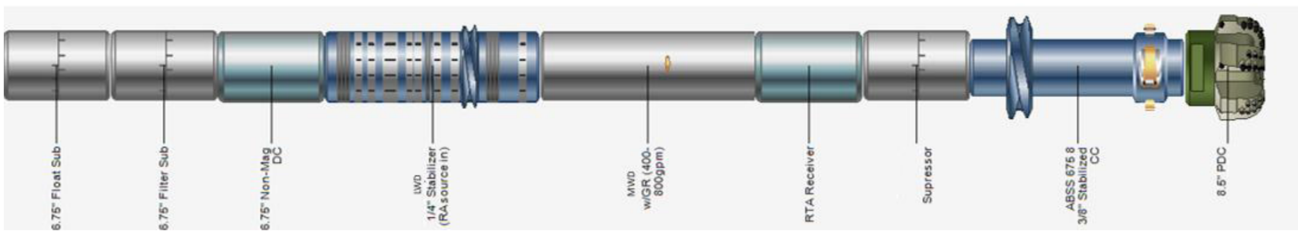

Figure 7—Case Study 2 BHA no 2: ABSS with Lithodensity Logging corrective BHA.

Results

The deployment of the ABSS system represented a substantial advancement over the previous 2nd Generation RSS run. It successfully met the aggressive trajectory requirements and mitigated the risk of a costly plug back or second sidetrack. In scenarios where the required dogleg severity (DLS) could not be achieved, the well trajectory would have bypassed the intended formation sub-layer—forcing the subsurface team to either compromise the reservoir target or initiate another sidetrack. By delivering precise curvature control, ABSS maintained alignment with the original geological objectives and ensured optimal reservoir placement.

This intervention not only corrected the initial trajectory deviation but also set a new standard for directional drilling efficiency in complex and heterogeneous formations.

High Directional Deliverability. The ABSS, featuring a piston bias unit design, consistently achieved a DLS of 5–6°/100 ft at 75% steering force (Table 4) throughout the critical build section. This enabled precise landing at the planned true vertical depth (TVD) and within the targeted lateral kickoff depth. The transition into the horizontal section was executed smoothly, and the wellbore condition remained excellent, supporting efficient progression through the lateral and minimizing risks associated with drag or borehole instability.

Beyond achieving the planned trajectory, the well also demonstrated excellent hole conditions. The drilling BHA was retrieved smoothly, with no signs of drag or tight spots—indicating minimal micro-doglegs and a clean, uniform borehole profile. This outcome highlights the effectiveness of the at-bit steering system in delivering not only precise directional control but also a high-quality wellbore, which is essential for efficient lateral drilling and optimized future completions.

Table 4—Case Study-2 BHA no 2: ABSS Deviation Survey

MD (ft)	Inc (°)	Azim Grid (°)	TVD (ft)	DLS (°/100ft)
9637.00	47.80	138.50	9486.48	4.44
9733.00	52.55	136.44	9547.95	5.21
9826.00	58.53	134.69	9600.55	6.61
9921.00	64.12	134.74	9646.12	5.88
10014.00	69.79	135.46	9682.51	6.14
10107.00	75.62	136.35	9710.14	6.33
10200.00	80.68	138.54	9729.24	5.91
10294.00	84.60	140.72	9741.28	4.76
10387.00	85.62	141.42	9749.21	1.33
10481.00	89.83	140.97	9752.94	4.50
10573.00	89.90	141.27	9753.15	0.33
10666.00	90.67	141.20	9752.69	0.83

Case Study 2 validated the performance gains observed in Case Study 1, with ABSS achieving a consistent DLS of 5–6°/100 ft under demanding conditions. The system enabled precise landing within the targeted sub-layer, eliminating the need for a second sidetrack. Its adaptive steering response ensured reliable trajectory control across heterogeneous formations. The success confirmed ABSS's capability to meet aggressive build-rate targets.

Conclusions and Recommendations

The successful deployment of the ABSS in operating company's unconventional project has demonstrated a significant advancement in directional drilling technology. The system consistently delivered high dogleg severity (DLS) up to 6°/100 ft at 75% steering force, outperforming conventional rotary steerable systems and eliminating the need for complex bottomhole assembly (BHA) modifications or secondary sidetracks. This capability was validated across both pilot and operationally critical runs, confirming ABSS's reliability in achieving aggressive build rates while maintaining trajectory precision.

Equally notable was the borehole quality maintained throughout the curve sections (Table 5). Despite the demanding trajectory requirements, the wells exhibited low friction factors (as low as 0.2), smooth BHA trips, and minimal micro-doglegs. These outcomes confirmed the tool's ability to drill high-curvature profiles without compromising wellbore integrity, resulting in enhanced drilling efficiency, reduced operational risk, and full preservation of reservoir access in line with geological objectives.

Table 5—Case Study-2 Comparative Performance: 2nd Generation RSS vs. ABSS

Feature	2nd Generation RSS	At Bit Steerable System ABSS
Formation Interaction	Struggled in heterogeneous formations due to inconsistent side force	Adaptive piston pressure modulation compensated for formation changes
Steering Response	Soft shale allowed pad penetration but reduced reactive force; hard shale resisted deflection, limiting DLS	Enhanced side force efficiency due to optimised bias unit geometry, maintaining DLS in both soft and hard layers
DLS Capability	Limited, especially in hard formations	Maintained 5–6°/100 ft consistently

Comparative Performance: 2nd Generation RSS vs. At Bit Steerable System ABSS

The ABSS system has transitioned from being formation-dependent to formation-dominant, consistently delivering required DLS regardless of lithology variations. Field data supports that the new bias unit design represents a generational leap in rotary steerable technology, particularly for complex unconventional wells requiring precise trajectory control in heterogeneous formations.

Looking forward, performance optimization will focus on evaluating ABSS's compatibility with alternative bit designs, especially less aggressive PDC bits, to improve tool reliability in high-shock, high-vibration environments. A tailored bit selection strategy, combined with vibration mitigation techniques, is expected to further enhance system performance and extend tool life.

Additionally, integrating this technology with autonomous drilling solutions—such as DD Advisor for downlink recommendations and Auto-downlinker for automated execution—can improve DLS delivery reliability while boosting overall drilling efficiency and wellbore quality in future deployments.

Terminologies

ABSS- At-the-Bit Steerable System
BHA- Bottomhole Assembly
DLS- Dogleg Severity (°/100 ft)
MD- Measured Depth
TVD- True Vertical Depth
RSS- Rotary Steerable System
MWD- Measurement While Drilling
ROP- Rate of Penetration
WOB- Weight on Bit

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